Dynamic Obstacle Detection and Prediction

Project plan

Author: Vladislav Stefanov

Fontys University of Applied Sciences

Eindhoven

March 16th, 2026

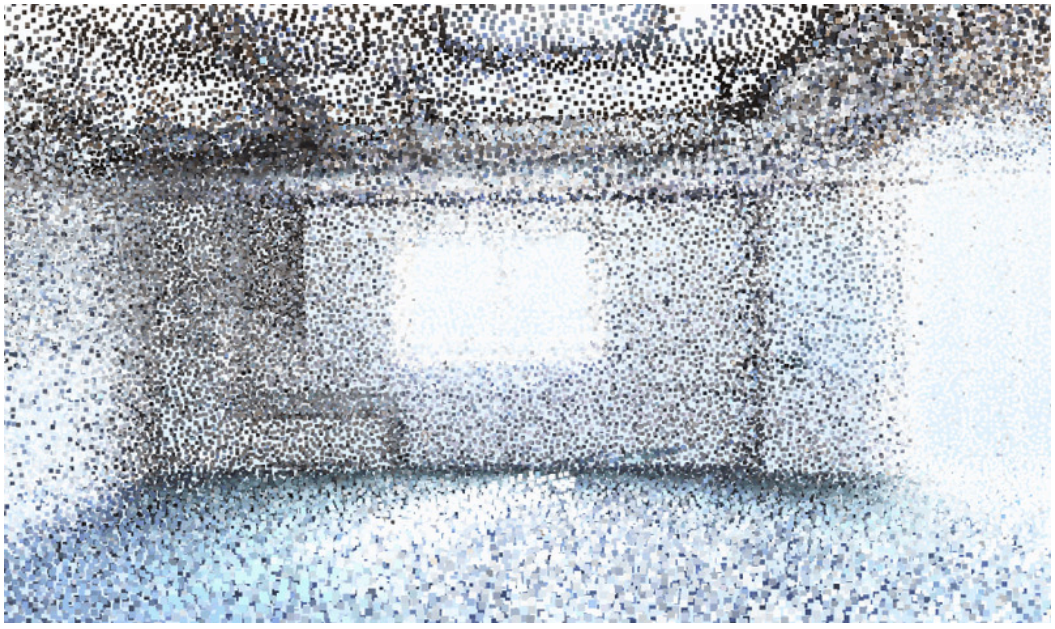


Table of Contents

1. 1. Context
2. 2. Problem Description
3. 3. Solution
    1. 3.1 Overview
    2. 3.2 Scope
        1. Phase 1 - Problem Understanding
        2. Phase 2 - State-of-the-Art Review and Experimental Exploration
        3. Phase 3 - Evaluation and Comparison
        4. Out-of-Scope (Current Definition)
    3. 3.3 Expected Contribution
4. 4. Pre-Conditions
5. 5. Research Questions
6. 6. Approach
7. 7. Learning Outcomes
8. 8. Time Plan
9. 9. Stakeholders and Communication
10. 10. Risk Management
11. 11. References

Version History

Version	Date	Ammendment
0.1	03.03.2026	First vesrion ready for review
1.0	16.03.2026	Applying feedback from stakeholders

1. Context

The graduation project is conducted within the High Tech Embedded Software (HTES) Research Group at Fontys ICT, part of Fontys University of Applied Sciences in Eindhoven, the Netherlands. HTES focuses on applied research in software methods for smart high-tech embedded systems, particularly in mobile robotics and cyber-physical systems..

Within the research group, multiple mobile robotics projects share a common technical limitation: the absence of a reliable system for tracking and motion prediction of dynamically moving obstacles based on 3D depth perception. Current projects rely on classical perception and filtering methods, while recent developments in AI-based world models have not yet been explored.

This project aims to design and evaluate a modular 3D perception and motion prediction pipeline, while also assessing whether world models offer practical advantages over established classical methods. **By addressing the current limitations, the project seeks to deliver a reusable perception module applicable across multiple HTES robotics platforms, including the Brum used in the Self-Driving Challenge, the Jackal UGV, the Boomba triangular robot platform, and the Avular Origin One.**

The results should provide a clear, evidence-based conclusion on whether world models represent a promising direction for further research within HTES or whether their practical constraints limit their suitability for mobile robotics applications.

2. Problem Description

Mobile robotic systems must perceive and interpret dynamic scenes in order to function safely and autonomously. While depth cameras are commonly used for collision prediction, the existing implementations within the HTES research context are limited in dynamic scenes.

The core problem is the absence of an object-centric perception pipeline that can:

1. Detect arbitrarily moving objects using depth data (detection).
2. Maintain consistent identities of those objects over time (tracking).
3. Predict their short-term future motion (prediction).

Without motion prediction, robotic systems react only to the present state of the environment. This reactive behavior limits performance in dynamic scenarios, particularly when obstacles move unpredictably. Higher-level modules such as navigation or path planning require not only knowledge of current obstacle positions, but also reliable short-horizon forecasts of their trajectories.

From a technical perspective, the problem involves several challenges:

- Depth data is noisy and incomplete.
- Object segmentation from point clouds must be robust to clutter and occlusions.
- Multi-object tracking requires reliable data association across frames.
- Motion prediction must account for uncertainty and changing motion patterns.
- The system must remain computationally feasible for real-time operation.

---

3. Solution

3.1 Overview

The final output of the project is a **modular 3D perception and motion prediction pipeline** that transforms raw depth measurements into:

1. Object-level **detections**
2. Stable object identities over time (multi-object **tracking**)
3. Short-term motion **predictions** with quantified uncertainty

The system must:

- Operate on 3D sensor data (depth data, LiDAR or RGB-D).
- Be modular, such that detection, tracking, and prediction components can be replaced or extended independently.
- Support real-time or near-real-time processing suitable for mobile robotics.
- Produce structured outputs usable by higher-level autonomy modules (e.g., navigation, planning).

A ROS2-based implementation is desirable for integration within HTES robotics projects, but the research and development of the perception pipeline is the primary requirement.

3.2 Scope

The plan is structured into three phases:

Phase 1 - Problem Understanding

- Understand robotics perception and analyze perception requirements for mobile robotics.
- Identify limitations of reactive obstacle detection systems.
- Define evaluation metrics.

Phase 2 - State-of-the-Art Review and Experimental Exploration

Topics to research, experiment with and evaluate:

Sensor Input

- Depth cameras
- RGB and RGB-D fusion
- LiDAR

Classical Methods

- Kalman Filters
- Multi-object tracking
- PCL (Point Cloud Library)
- Open3D
- CUDA acceleration for point cloud processing
- Physics-based game engines for simulation

State-of-the-Art

- ZED SDK (highest priority)
- Structured motion representations (SMORE)
- World models for dynamic environments
- NVIDIA Cosmos
- MONet
- Dreamer v3
- DynamicNeRF
- HexPlane
- Autoware
- OpenUSD
- Depth Anything v3
- Isaac Perceptor
- Dino v2
- ConceptGraphs
- ...

Practical and Educational Resources

- NVIDIA free online robotics and computer vision courses
- Self-driving challenge documentation
- Robotics perception pipelines

If new relevant approaches are discovered during the project, they may be incorporated following evaluation of their relevance and feasibility.

Phase 3 - Evaluation and Comparison

Evaluation metrics must be explicitly defined and justified. Candidate evaluation criteria:

- Accuracy (task correctness)
- Robustness (performance under harder scenarios)
- Computational efficiency (latency, throughput)
- Stability (consistency)
- Scalability (number of obstacles)

Out-of-Scope (Current Definition)

- Robot motion planning
- Full autonomous system deployment
- Long-term behavior prediction (>5 seconds)
- Large-scale deep learning model training (unless feasible within timeline)

3.3 Expected Contribution

The project is expected to contribute:

1. A modular perception pipeline for dynamic obstacle tracking and prediction.
2. Empirical comparison of classical methods vs world-model approaches.
3. Experimental validation under controlled conditions.
4. A reusable foundation for HTES mobile robotics projects.

---

4. Pre-Conditions

- Working Agile using a Git issue board.
- Using GitLab for version control.
- Using the compute resources of the bigboi PC.
- Using the ISSD for hardware resources (e.g. depth camera).

---

5. Research Questions

Main research question:

How can a modular 3D depth-based perception system reliably detect, track and predict the motion of dynamic obstacles in mobile robotics?

Sub-questions:

- What are the technical problems with depth-based perception and why are they problems?
- What technologies solving similar problems are available (both classical and state-of-the-art)?
- How to implement one of the available methods and investigate its capabilities?
- What evaluation criteria/metrics are suitable for comparison between different approaches?

6. Approach

During the internship, I will be using the Agile Scrum approach working with a Git issue board and 2- or 3-week-long sprints. During each sprint, I will perform a research-implement-test cycle for different technologies, sharing the results with the stakeholders. The decision what to work on during a sprint will depend on the stakeholders and will be decided at the beginning of the sprint. We will have weekly meetings with the company mentor and the product owner and another meeting with the first assessor to update them on my progress and ask for feedback.

7. Learning Outcomes

Learning outcome	**Evidences**
Learning Outcome 1: Professional Duties *You create professional products at bachelor's level within your IT subdomain, in line with the standards and working methods customary in the sector, and you will act accordingly in relevant complex professional situations.*	- project plan - research documentation - experiments and demos of different approaches - comparison between different approaches - prototype of a 3D-depth perception pipeline
Learning Outcome 2: Situation-Orientation *You apply your knowledge and skills in an authentic professional context, where your activities and choices are aligned with the*	- feedback from the stakeholders - demos

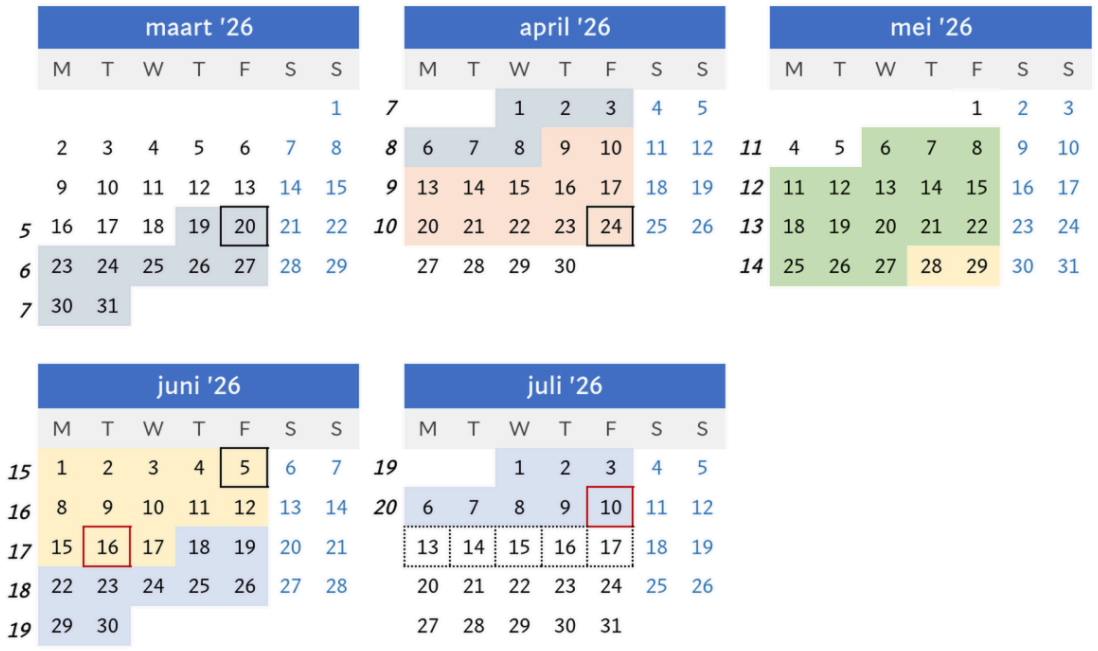
needs, processes and standards of the organisation and lead to demonstrable value for stakeholders.	
Learning Outcome 3: Professional Standards *You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide advice to your stakeholders even in complex and uncertain contexts. You substantiate and validate future-oriented choices by use of ethical, intercultural and sustainable arguments.*	- research documentation - demos
Learning Outcome 4: Personal Leadership *You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show that you are capable of carrying out these actions and achieving your goals, adjusting them as necessary.*	- project plan - Git issue board - feedback from the stakeholders

8. Time Plan

The internship planning is split in five sprints with a length of three weeks on average, followed by the graduation conference. Additionally, there are four portfolio submissions in total I need to provide evidences for.

Sprint N	Duration
Sprint 1	Mar 19 - Apr 8
Sprint 2	Apr 9 - Apr 24
Sprint 3	May 6 - May 27
Sprint 4	May 28 - Jun 17
Sprint 5	Jun 18 - Jul 10
Graduation conference	Jul 13 - Jul 17

Portfolio Submission	Date
Portfolio snapshot 1 including project plan (week 5)	March 20th
Portfolio snapshot for midterm review (week 10)	April 24th
Concept of final portfolio (week 15)	June 5th
Final portfolio submission (week 17)	June 17th



9. Stakeholders and Communication

Name	Email	Role and Responsibilities	Availability
Vladislav Stefanov	v.stefanov@student.fontys.nl	Intern Executes and coordinates the project	Every day during the working week
Edwin van den Oetelaart	e.vandenoetelaar@fontys.nl	Company mentor	Every Wednesday, 15:00-16:00 Online or in person during the working week
Martin Roa Villescas	m.roavillescas@fontys.nl	Product owner	Every Wednesday, 15:00-16:00 Online during the working week
Teade Punter	teade.punter@fontys.nl	Research chair of HTES	Once in several weeks on

			Wednesday, 15:00-16:00 Online during the working week
Mark Beks	m.beks@fontys.nl	First assessor	Every Monday, 13:30-14:00 Online during the working week
Ben Schreur	b.schreur@fontys.nl	Second assessor	Midterm review, final presentation

10. Risk Management

Risk Category	Description	Probability	Impact	Mitigation Strategy
Scope creep	Expansion into too many advanced methods (e.g., world models, deep learning) exceeding available time.	High	High	- Define strict MVP - Frequent feedback sessions with stakeholders - Perform experiments, not only research
Technical complexity	Selected approaches (e.g., tracking, prediction models) prove more complex than expected.	Medium	High	- Start with simple baseline models - Incrementally increase complexity
Evaluation ambiguity	Metrics are poorly defined, leading to unclear or non-comparable results.	Medium	High	- Define evaluation metrics early - Use established benchmarks where possible
Time constraints	Underestimation of development or experimentation time.	Medium	High	- Include buffer time

				- Work Agile using issues and sprints
Knowledge gaps	Insufficient theoretical understanding of filtering, perception, or motion modelling.	Medium	Medium	- Structured literature review - Targeted courses (e.g., NVIDIA robotics/vision) - Experimentation

11. References

Scrum Guide | Scrum guides. (n.d.). https://scrumguides.org/scrum-guide.html